

Global Air Pollution -Removal of Outliers and their impact on model

IQR Method

Step 1: Import Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

- pandas → Used for data loading and manipulation.
- numpy → Used for numerical operations.
- matplotlib.pyplot → Used for creating visualizations.

Step 2: Load the Dataset

- `pd.read_csv()` → Reads a CSV file and converts it into a Pandas DataFrame.
- `df` → Stores the dataset for further analysis.

Step 3: Calculate Quartiles

`quantile(0.25)`- Finds the first quartile (Q1).

`quantile(0.50)`-Finds the second quartile (Q2), or median.

`quantile(0.75)`-Finds the third quartile (Q3).

Q1 →25th percentile.

Q2 →50th percentile.

Q3 →75th percentile.

Step 4: Calculate IQR

$IQR = Q3 - Q1$

Step 5: Calculate Outlier Limits

$Q1 - 1.5 * IQR$ → Calculates the lower boundary for potential outliers.

$Q3 + 1.5 * IQR$ → Calculates the upper boundary for potential outliers.

`lower_limit` → Values below this are potential outliers.

`upper_limit` → Values above this are potential outliers.

Formula:

Lower Limit= $Q1 - 1.5(IQR)$

Upper Limit= $Q3 + 1.5(IQR)$

Step 6: Count the Outliers

- `df[...]` → Filters the DataFrame based on the given conditions.
- `< lower_limit` → Finds values below the lower boundary.
- `upper_limit` → Finds values above the upper boundary.
- `|` → Represents OR; either condition can identify an outlier.
- `len(outliers)` → Counts the number of detected outlier observations.

The IQR method identified 2,641 potential PM2.5 outliers out of 23,463 observations.

Using the IQR method, 2,641 observations were identified as potential outliers in the PM2.5 AQI Value, representing approximately 11.26% of the dataset.

Step 7: Remove PM2.5 Outliers Using IQR

- `>= lower_limit` → Keeps values within the lower boundary.
- `<= upper_limit` → Keeps values within the upper boundary.
- `&` → Requires both conditions to be true.
- `.copy()` → Creates an independent copy of the filtered dataset.
- `df_iqr` → Stores the dataset after IQR-based outlier removal.

The IQR method identified 2,641 potential outliers in PM2.5 AQI Value. These observations were excluded to create a separate dataset for the improved regression model. The original dataset was retained so that the performance of the original and outlier-treated models could be compared fairly.

Conclusion:

After applying IQR-based outlier treatment to PM2.5 AQI Value, MAE, MSE, and RMSE decreased, indicating lower prediction errors. However, the R^2 score decreased from 0.9694 to 0.9033. Therefore, IQR-based outlier removal cannot be considered an overall improvement based on the current comparison. Further investigation using a consistent train-test split and an alternative outlier-detection method such as the 3-sigma technique is required.

Winsorization Technique

Winsorization is an outlier-treatment technique in which extreme values are replaced with predefined boundary values instead of being removed.

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Step 2: Load the Dataset

- `pd.read_csv()` → Reads a CSV file and converts it into a Pandas DataFrame.
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Step 3: Calculate Quartiles

`quantile(0.25)`- Finds the first quartile (Q1).

`quantile(0.50)`-Finds the second quartile (Q2), or median.

`quantile(0.75)`-Finds the third quartile (Q3).

Q1 → 25th percentile.

Q2 → 50th percentile.

Q3 → 75th percentile.

Step 4: Calculate IQR

$IQR = Q3 - Q1$

Step 5: Calculate Outlier Limits

$Q1 - 1.5 * IQR$ → Calculates the lower boundary for potential outliers.

$Q3 + 1.5 * IQR$ → Calculates the upper boundary for potential outliers.

`lower_limit` → Values below this are potential outliers.

`upper_limit` → Values above this are potential outliers.

Formula:

Lower Limit= $Q1 - 1.5(IQR)$

Upper Limit= $Q3 + 1.5(IQR)$

Step 6: Replace Outliers Using Winsorization

Instead of removing the observations identified as outliers, Winsorization replaces them with the nearest acceptable boundary value.

- `df.copy()` → Creates a separate copy of the original dataset.
- `.clip()` → Replaces values outside the specified limits with the corresponding boundary values.
- `lower=lower_limit` → Values below the lower limit are replaced by the lower limit.
- `upper=upper_limit` → Values above the upper limit are replaced by the upper limit.

Conclusion:

Winsorization was applied to the PM2.5 AQI Value to reduce the influence of extreme observations while retaining all data points. However, after applying Winsorization, the R^2 score decreased to approximately 0.72 compared with the original R^2 score of 0.9694. Therefore, Winsorization did not improve the regression model and was not selected as the preferred outlier-treatment technique.

3 sigma

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Step 2: Load the Dataset

- pd.read_csv() → Reads a CSV file and converts it into a Pandas DataFrame.
- df → Stores the dataset for further analysis.

Step 3: Calculate Mean, Median and Standard deviation

.mean() → Calculates the arithmetic mean of PM2.5 AQI values.

.median() → Finds the middle value after arranging the observations in order.

.std() → Measures how spread out the observations are from the mean.

Step 4: Outlier detection

For a variable with mean and standard deviation :

Therefore, any observation satisfying:

or

is considered a potential outlier.

Conclusion:

The 3-sigma method was applied to detect and remove extreme PM2.5 AQI Value observations. It identified 288 potential outliers, which was considerably fewer than the 2,641 outliers identified using the IQR method. After removing these observations, MAE, MSE, and RMSE decreased, while the R^2 score remained high at 0.9599 compared with the original 0.9694. Therefore, the 3-sigma method provided a better balance between reducing prediction error and retaining the model's explanatory power compared with IQR-based outlier removal.

Final Conclusion

The 3-sigma technique was applied to the PM2.5 AQI Value to identify extreme observations using the mean and standard deviation. The method identified 288 potential outliers, which was considerably lower than the 2,641 outliers identified using the IQR method.

After removing the 288 observations identified by the 3-sigma method, the regression model showed improved error metrics. MAE decreased from 5.616 to 4.839, while RMSE decreased from 9.877 to 8.562. This indicates that the model's prediction errors were reduced after the outlier treatment.

However, the R^2 score decreased slightly from 0.9694 to 0.9599. Therefore, the 3-sigma method did not improve the R^2 score, but it provided a good balance between reducing prediction errors and retaining a strong relationship between PM2.5 AQI Value and AQI Value. Compared with the IQR method, 3-sigma produced a much stronger final model.